

UNIT-11

$$1. \frac{f}{p} = \frac{f}{mv}$$

↳ Black body radⁿ.

↳ Statistical Mechanics

↳ Quantum Mechanics.

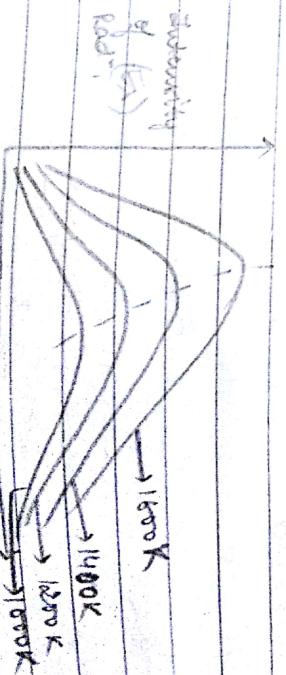
Classical - X $\rightarrow$ only macroscopic obj.
Quantum - X $\rightarrow$ size, number, laws

* Black body radⁿ:

A black body is defined as one which abs. the total radⁿ. of all wavelengths incident on it.
eg: lamp black $\neq$ Pt black.

• Variation of intensity of black body radⁿ:-

When a black body is heated at diff. temp., it emits radⁿ. of all types of radⁿ. The origin of radⁿ. from a heated body is the rapid vibrating particles known as oscillators. Acc. to Maxwell's EM theory, these oscillators emit radiant energy in the form of EM waves.
The variation of intensity of radⁿ. at diff. temp is explained by the figure:-



2M

the foll. conclusions can be drawn from the fig.

- (1) At a given temp., the energy isn't uniformly distributed in the radⁿ. spectrum of hot body.
- (2) At a given temp., intensity of given radⁿ. $\uparrow$ with $\uparrow$ in λ $\neq$ at a particular λ , its value is max. With further $\uparrow$ in λ , the intensity of heat radⁿ. $\downarrow$.

- (3) With $\uparrow$ in temp., $\lambda_m \downarrow$ such that:

$$\lambda_m T = \text{const} = 0.2896 \text{ cmK}$$

(where λ_m is the λ at which max. emission of Energy takes place)

$\Rightarrow$ This relⁿ. is called WEIN'S DIST. LAW.

- (4) For all λ s an $\uparrow$ in temp., causes an $\uparrow$ in energy emission.
- (5) Stephen's law is satisfied by all the temp. graphs acc. to which the intensity of total radⁿ. is $\propto$ the fourth power of absolute temp.

WEIN'S DISTRIBⁿ. LAW -

Acc. to Wein found that the intensity of radⁿ. is the funcⁿ of λ $\neq$ the energy dens. of black body radⁿ. of λ s b/w λ & $\lambda + d\lambda$ for the isothermal cavity of temp. T is given by

$$E_\lambda d\lambda = \frac{A}{\lambda^5} e^{-B/\lambda T} d\lambda$$

$\times$ cancel 'integer' form

A, B - are const.

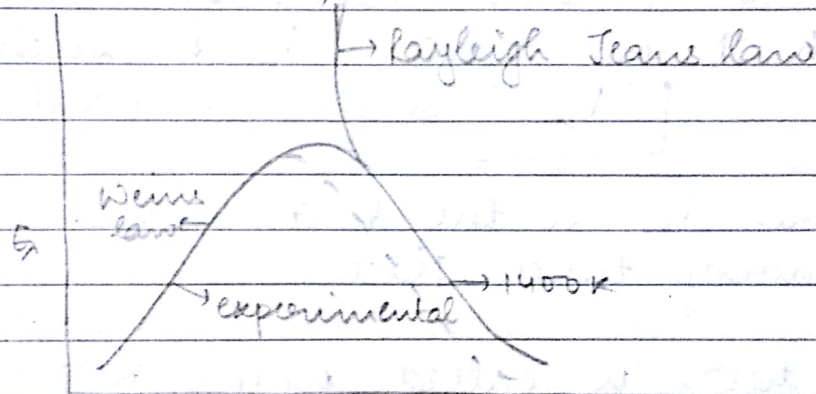
$\Rightarrow$ applicable to lower λ only.

• RAYLEIGH JEANS DISTRIBⁿ. LAW -

$$E_{\lambda} d\lambda = \frac{8\pi^5}{15} k_b T d\lambda \rightarrow \text{applicable to higher } \lambda \text{ only}$$

↳ const.

• ULTRA-VIOLET CATASTROPHE / (Failure of classical theory to explain spectral distrⁿ. of energy)



It's obs. that Weins 5th power law agrees with exptal. results only in the lower λ region, while Rayleigh Jeans law on the other hand can account for the distrⁿ. of energy in the higher λ & region & fails completely in shorter region. This result is said to be UV catastrophe.

* The discrepancies b/w. of black body radⁿ. is called UV catastrophe / Jeans paradox.

* Planck's Quantum Theory of Black Body Radⁿ. :-
 On the failure of Rayleigh Jeans Law to explain spectral & energy curve of black body radⁿ.
 Max Planck put forward the quantum theory to explain the experimentally obs. spectrum of black body radⁿ. He suggested that the correct results can be obt. if the energy of oscillating c- is taken as discrete. He derived the radⁿ. law using some assumptions:

- (1) A chamber containing black body radⁿ. also contains simple harmonic oscillators of molecular dimensions which can vibrate with all possible λ s.
- (2) The λ of radⁿ. emitted by an oscillator is same as the λ of its vibrations.
- (3) An oscillator can't emit energy in a continuous manner while in multiples of small unit ~~called~~ called quanta of magnitude $h\nu$.
- (4) Oscillators can emit or abs. radⁿ. energy in packets of magnitude $h\nu$. $\Rightarrow$ Exchange of energy b/w radⁿ. & matter isn't continuous.

If N be the no. of Planck's Oscillators & E their energy (each), then $\langle E \rangle / \text{oscillator} = \bar{E}$.

$$\bar{E} = \frac{E}{N} \quad \text{--- (1)}$$

Let there be $N_0, N_1, N_2, \dots, N_n$ oscillators having energy $0, E', 2E', \dots, nE'$ resp.

Now we've $N = N_0 + N_1 + N_2 + \dots + N_n$ — (2) $\neq$

$$E = E_0 + E'_1 + 2E'_2 + \dots + nE'_n \quad (3)$$

$\therefore (E = n h \nu)$

Acc. to Maxwell's distrib. formula, no. of oscillators having energy nE' is given by,

$$N_n = N_0 e^{-nE'/k_B T} = N_0 \exp\left(\frac{-nE'}{k_B T}\right) \quad (4)$$

Subst^y the values of $N_1, N_2, N_3 \neq$ so on from eqⁿ (4) in eqⁿ (2) $\neq$ eqⁿ (3).

$$N = N_0 + N_0 \exp\left(\frac{-E'}{k_B T}\right) + N_0 \left\{ \exp\left(\frac{-2E'}{k_B T}\right) + \dots + N_0 \exp\left(\frac{-nE'}{k_B T}\right) \right\}$$

$$= N_0 \left[1 + \exp\left(\frac{-E'}{k_B T}\right) + \exp\left(\frac{-2E'}{k_B T}\right) + \dots + \exp\left(\frac{-nE'}{k_B T}\right) \right]$$

Let us assume that : $\exp\left(\frac{-E'}{k_B T}\right) = x$

$$N = N_0 [1 + x + x^2 + \dots]$$

$$= N_0 \frac{1}{(1-x)}$$

$$N = \frac{N_0}{1 - \exp\left(\frac{-E'}{k_B T}\right)}$$

Putting value of N in eqⁿ (3)

$$E = N_0 \times 0 + E' N_0 \exp\left(\frac{-E'}{k_B T}\right) + E' N_0 \exp\left(\frac{-2E'}{k_B T}\right) + \dots + 2E' N_0 \exp\left(\frac{-2E'}{k_B T}\right) + \dots$$

$$= N_0 E' \exp\left(\frac{-E'}{k_B T}\right) \left[1 + 2 \exp\left(\frac{-E'}{k_B T}\right) + 3 \exp\left(\frac{-2E'}{k_B T}\right) + \dots + 2 \exp\left(\frac{-(n-1)E'}{k_B T}\right) \right]$$

Let $\exp\left(\frac{-E'}{k_B T}\right) = x$

$$E = N_0 E' x \left[1 + 2x + 3x^2 + \dots + 2x^{n-1} \right]$$

$$= \frac{N_0 E' \exp\left(\frac{-E'}{k_B T}\right)}{(1-x)^2}$$

$$= N_0 E' \exp\left(\frac{-E'}{k_B T}\right) \quad \text{--- (6)}$$

$$\left[1 - \exp\left(\frac{-E'}{k_B T}\right) \right]^2$$

Subst^y value of $E' \neq N$ from eqⁿ (5), (6) in eqⁿ (1)

$$\bar{E} = \frac{E}{N} = N_0 E' \exp\left(\frac{-E'}{k_B T}\right) \left[1 - \exp\left(\frac{-E'}{k_B T}\right) \right]$$

$$\left[1 - \exp\left(\frac{-E'}{k_B T}\right) \right]^2 \times N_0$$

$$= E' \exp\left(\frac{-E'}{k_B T}\right)$$

$$\exp\left(\frac{-E'}{k_B T}\right) \left[\exp\left(\frac{+E'}{k_B T}\right) - 1 \right]$$

$$\bar{E} = \frac{E'}{\left[\exp\left(\frac{+E'}{k_B T}\right) - 1 \right]} = \frac{h\nu}{\left[\exp\left(\frac{+h\nu}{k_B T}\right) - 1 \right]} \quad \text{--- (7)}$$

10M

Eq. (7) represents the avg. energy of Planck's oscillator. Now we know that the no. of oscillators/unit vol. in ν range $\nu \pm (\nu + d\nu)$ can be written as:

Learn formula from

$$N = N(\nu) d\nu = \frac{8\pi \nu^2 d\nu}{c^3} \quad (8)$$

⑧ $\times$ ⑦ we get the total Energy/unit vol. belonging to range $d\nu$.

Experi. of Planck's law of ν

$$E(\nu) d\nu = \frac{8\pi h \nu^3}{c^3 [\exp(\frac{h\nu}{k_b T}) - 1]} d\nu \quad (9)$$

represents internal. $\times$ cancel

$$\cancel{E(\nu) d\nu} = \cancel{8\pi h \nu^3} \quad \nu = \frac{c}{\lambda} \quad \therefore d\nu = \left| \frac{-c d\lambda}{\lambda^2} \right|$$

$$\frac{c}{\lambda^2} E(\lambda) d\lambda = \frac{8\pi h c}{\lambda^5 [\exp(\frac{hc}{\lambda k_b T}) - 1]} \frac{c d\lambda}{\lambda^2}$$

part of E_{expt}

$$E(\lambda) d\lambda = \frac{8\pi h c}{\lambda^5 [\exp(\frac{hc}{\lambda k_b T}) - 1]} d\lambda \quad (10)$$

* Derive both laws with help of Planck's law:
Wien's law - for low λ
- λT is small

$$\Rightarrow \exp\left(\frac{hc}{\lambda k_B T}\right) > 1$$

On neglecting 1 in eqⁿ (10)

$$\boxed{E_{\lambda} d\lambda = \frac{8\pi hc}{\lambda^5} \exp\left(-\frac{hc}{\lambda k_B T}\right) d\lambda} \quad (11)$$

Rayleigh Jeans law - for high λ

$\rightarrow \lambda T$ is large

$$\Rightarrow \exp\left(\frac{hc}{\lambda k_B T}\right) = 1 + \frac{hc}{\lambda k_B T} + \frac{1}{2} \left(\frac{hc}{\lambda k_B T}\right)^2 + \dots$$

expand.

Neglecting higher terms we get,

$$\exp\left(\frac{hc}{\lambda k_B T}\right) \approx 1 + \frac{hc}{\lambda k_B T}$$

$$\therefore \boxed{E_{\lambda} d\lambda = \frac{8\pi k_B T d\lambda}{\lambda^4}} \quad (12)$$

* Statistical Mechanics: (both classical & quantum)

$\rightarrow$ Follows Newton's laws (for macroscopic obj. i.e. $> 10^{-6}$ m)

Imp.

- Diff b/w sym. & antisym.

• + sym

- anti

• bosons

fermions

• Bohr-Einstein st. • Fermi-Dirac statistics

eg. photons/phon. eg. e^- , p^+

- Diff b/w bosons & fermions

V. Imp 100%

Phase space:

space where position of particle can be defined using 6 coord.

Last 2 from above

sym & antisym

bosons - integral spin
X follow Pauli Excl. Pr.

fermions - half spin

Diff. b/w 3 statistical distr.

(g, gas mol.) Maxwell Boltzmann	(ph, ph) Bohr-Einstein	(e ⁻ , p ⁺) Fermi-Dirac
1) Identical distinguishable particles (nature) large size	1) Identical, indistinguishable	1) Identical, indistinguishable
2) Fixed (X defined) (size of phase cell)	2) Defined = h^3	2) Defined = h^3
3) Continuous distr. (no. of phase cells) X defined	3) Can be div. into 3) (etc.) no. of cells. It can't be $> h^3$ (vol. of p.c.)	3) " "
4) Any no. of particles can be defined	4) In single p.c. pairs of particles exist	4) 1 pc - 1 particle + - (diff. cells)
5) Macrostates can have > 1 microstates	5) Each macro = 1 micro (E levels) (discrete)	5) " "
6) Continuous E distr.	6) Discrete	6) " "
7) Spinless particles	7) Integral spin	7) Half spin
8) X P. exclusion pr.	8) X	8) Strictly follow
9) Occup ⁿ index (distr. of particles in E levels) $n_i = \frac{g_i}{e^{K+\beta E_i}}$	9) $n_i = \frac{g_i}{e^{K+\beta E_i} + 1}$	9) $n_i = \frac{g_i}{e^{K+\beta E_i} - 1}$
10) X exist for gas mol.	10) sym ψ	10) antisym ψ (-) coord. change

→ Constraints:

① No. of particles $\sum n_i$ should be const.

* Maxwell-Boltzmann distribⁿ law:-

Let us consider a sys. of N particles distinguishable molecules of gas. Suppose n_1 of them have E_1 , n_2 have E_2 , ... n_i have E_i .

This entire assembly of molecules can be divided into diff. energy states, so 2 constraints are to be imposed: 1) total no. of molecules is const.

$$N = n_1 + n_2 + n_3 + \dots + n_i + \dots$$

$$N = \sum_{i=1}^{\infty} n_i$$

const. diff. $\rightarrow \sum_{i=1}^{\infty} n_i = N$ — (1)

2) total energy of gas molecules is constant

$$E = n_1 E_1 + n_2 E_2 + \dots + n_i E_i + \dots$$

$$E = \sum_{i=1}^{\infty} n_i E_i$$

$$\sum_{i=1}^{\infty} E_i n_i = E$$
 — (2)

$\rightarrow \because E$ of each cell = const.

Suppose there are g_i cells of energy E_i . Total no. of ways in which n_i molecules have energy E_i is $(g_i)^{n_i}$. Hence total no. of ways in which N molecules can be distributed among various energies is:

$$\sum \rightarrow \prod$$

$$\text{multiply} = \prod$$

$$\omega_1 (g_1)^{n_1} (g_2)^{n_2} \dots (g_i)^{n_i} \quad (3)$$

The no. of ways in which the gps. of $n_1, n_2, \dots, n_i$ can be particles can be chosen from N particles is given by:

$$\omega_2 = \frac{N!}{n_1! n_2! n_3! \dots} \quad (4)$$

$\therefore$ No. of distinguishable ways in which M molecules can be distributed among the possible energy levels $\Rightarrow$

$$\omega = \omega_1 \omega_2 = \frac{N!}{n_1! n_2! n_3! \dots} (g_1)^{n_1} (g_2)^{n_2} (g_3)^{n_3} \dots \quad (5)$$

Applying log to generalized cond. of eq. (5)

$$\omega = \frac{N!}{\prod n_i!} \prod (g_i)^{n_i}$$

log (multiply $\rightarrow$ add)

$$\ln \omega = \ln N! + \sum_{i=1}^s \ln (g_i)^{n_i} - \sum_{i=1}^s \ln (n_i!) \quad (6)$$

$$= \ln N! + \sum_{i=1}^s n_i \ln g_i - \sum_{i=1}^s n_i \ln n_i$$

$$[\ln x! = x \ln x - x]$$

Applying Stirling th^o to eq. (6)

$$\ln \omega = \ln N!$$

$$\ln \omega = N \ln N - N + \dots$$

$$\ln \omega = N \ln N - N - \sum [n_i \ln n_i - n_i] + \sum [n_i \ln g_i]$$

$$= N \ln N - N - \sum n_i \ln n_i - \sum n_i + \sum (n_i \ln g_i) \quad (7)$$

We have max. value of ω . For max. ω diff.ⁿ of left term will be ≥ 0 .

$\rightarrow N$ const, $\therefore$ diff.ⁿ $\rightarrow 0$.

$$\delta (\ln \omega)_{\max} = - \sum \left(n_i \times \frac{1}{n_i} \delta n_i \right) - \sum (\ln n_i \delta n_i) + \sum (\delta n_i \ln g_i) = 0$$

prob. = const.

$$- \sum [\ln(n_i)] \delta n_i + \sum [\ln(g_i)] \delta n_i = 0 \quad (8)$$

Multiply eqⁿ. ① with $-\alpha$ & eqⁿ. ② with $-\beta$ & adding these 2 products in eqⁿ. ⑧

$$\rightarrow \sum (-\ln(n_i) + \ln(g_i) - \alpha - \beta E_i) \delta n_i = 0$$

~~$\neq 0$ from eqⁿ ①~~

$\delta n_i = 0$ from eqⁿ ①

$$\therefore -\ln(n_i) + \ln(g_i) - \alpha - \beta E_i = 0$$

$$g_i / n_i = e^{-(\alpha + \beta E_i)}$$

$$\boxed{n_i = \frac{g_i}{e^{\alpha + \beta E_i}}}$$

$$\beta = \frac{1}{k_B T}$$

$$\alpha = -\frac{\mu}{k_B T} \rightarrow \text{chemical potential}$$

100%

grad, div., curl — def.ⁿ / numerical
eqⁿ. of continuity — phy — sig.
integral form — " "
dielectric
Planck's radⁿ. law
table